University of Essex

Online

Reflection on Research Methods and Professional Practice Module

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Introduction (What?)

The module, Research Methods and Professional Practice, has been the scaffold on which my MSc experience has been built to help me shift my mindset towards a more rigorous academic researcher, as opposed to a technical practitioner. The final product of this process is a proposal to research an AI-optimised IoT framework to deal with water scarcity in the UAE. The three main areas that are critically reflected in this reflection are how I have developed my skills in statistical analysis and its application in the field of academic research, the iterative nature of the process, and the overall significant influence of the process on my professional identity and ethics. In order to organise this examination, I will use the model of What? So What? and Now What? developed by Rolfe et al. (2001).

Statistical Analysis: From Intimidation to Application (So What?)

At first, the statistics of Units 6-9 were a cause of anxiety. Being an AI student, I was at ease when using machine learning models but not when mathematically vindicating them. The completion of the required worksheets on descriptive and inferential statistics constituted a momentous experience of So What? Computation of confidence intervals of simulated data of the water usage or the hypothesis test takes statistics out of the scholarly speculation stage and into the utility phase of assuring robustness.

This was very vital to my research proposal. It changed my assessment plan, which was a general direction of testing the model, into a particular protocol with quantitative measures such as F1-Scores and the strict cross-validation k-fold (University of Edinburgh, n.d.). Now I realise I was not being statistically literate, but it is a required standard when conducting credible AI research. It is the language that shows that the efficacy of my model is not an exception but a statistically significant result. This has taught me directly how to make my proposal more rigorous and defendable, to make my validation of the AI artefact be founded on empirical principles.

The Research Process: Embracing Iteration and Criticality (So What?)

The progressive path between the general research interest in AI sustainability and a targeted research proposal was painful and challenging. The first significant obstacle was the literature review. My early efforts were summative, where I described papers instead of synthesising them in a bid to realise a gap. This lack of criticality was pointed out in feedback. Referring to the literature again, I was on the hunt to find the contradictions and limitations, which brought me to the obvious gap that researchers such as Naqash et al. (2023) have recognised: the lack of connection between IoT data collection and intelligent action.

My critical lens was formed basically around this lens of So What? It made me have to put in defence not only what I was undertaking, but why it was a necessary contribution. The fact that I decided to use Design Science Research (DSR) as my methodology was a consequence of this process- it was the most fitting methodology that would be used to construct and test a new artefact. Moreover, I had to leave the technical specifications because of modules such as ethics (Unit 1) and professional issues. I was forced to think critically about possible biases in my AI models that would result in unfair distribution of water and take into account the social effects on the workforce. This holism of the technical work that cannot be seen out of its context of ethical and social considerations has been among the greatest changes in my thought.

Professional Development: Integrating Ethics and Forward Planning (Now What?)

The professional skills matrix and SWOT analysis made me have some form of organised Now What? for my career. It also clarified that though I am strong in my technical skills in Python and ML frameworks, I have a weakness in my communication skills in conveying complex research to various audiences and handling big projects. My action plan, according to which I have concentrated on enhancing academic writing and pre-emptive risk planning, is already being put into practice. Having created the Gantt chart and risk register for my proposal (Unit 12) was a good experience in managing a project, which involved converting the research plan into a manageable timeline with reduced risks, such as data unavailability.

The attention to professional codes, such as the code of conduct of BCS and the ACM code of conduct, in the module has permanently transformed my professional structure. Ethics was considered to me earlier as a box to be checked. I now regard it as a part of the design process. Data privacy and algorithmic fairness are not the things that my proposal comes up with in a later discussion, but are the design principles. It is an attitude that I am going to bring into all my future engagements, where my work as an AI professional will be guided by the principles of accountability and social good.

Conclusion

This critical experience highlights the fact that research is not a straight journey but a learning, application and evaluation cycle. The module has not only taught me a set of methods, but also taught me an attitude of strict, ethical and reflective practice. The statistical abilities provide me with the means to justify, the research methodology provides me with the framework to investigate, and the career growth provides me with the accountability to act in an upright manner. Being both an aspiring research-informed professional and a student of AI, I am not only an intelligent student of AI but a responsible and impactful one, ready to provide solutions as I advance to my dissertation and my future professional life. It is a continuous promise to use this integrated learning to solve real-world problems, beginning with the sustainable operations of the valuable water resources in my country.

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